PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

SERIES 1 • CARD 07 / 12 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS

CARD 07 / 12 • PROB-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.

Rolling two fair 6-sided dice gives 36 outcomes:

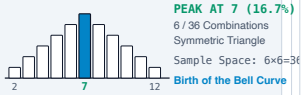
Sample Space = 6 × 6 = 36 Pair Outcomes

2d6 sum forms a triangle distribution peak at 7!

CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/



PEAK AT 7 (16.7%)

6 / 36 Combinations

Symmetric Triangle

Sample Space: 6×6=36

Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC	COLOR
Fair d6 Dice	4x	16.0 mm Fair	High-Contrast
Polyhedral d20	1x	20-Sided d20	Emerald Green
Precision Hub	1x	Brass Hub	High-Spin Brass
Spinner Dials	4x	3 Dials Set	Polymer Plates
Sample Grid Lid	1x	6×6 Matrix	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Seat the 4 d6 dice and 1 d20 into bottom bays. Stack spinner dials flat and click brass hub onto post. Empty bays show missing pieces instantly. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The sum of independent random variables tends toward a normal distribution as sample size grows.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

SERIES 1 • CARD 07 OF 12 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS

CARD 07 / 12 • PROB-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/dice-and-spinner/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 7.SP.C.8 (Compound Events)
- CCSS HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representati...)

Grades 5–10 Probability & Statistics

GUIDING INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

① STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1-6) × Die 2 columns (1-6) = 36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

② STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:
- Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
- Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P =

③-④ STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.

4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:

Press Dice & Statistics Icon ([📊], [Σ+] & [g][x̄]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.

Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60

⑤ CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together,
extreme sums require extreme coordination, while central sums have many different additive combinations!
This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

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STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PROB-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:

Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 14 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California